

FORM TP 2013021

TEST CODE **01238010**

JANUARY 2013

CARIBBEAN EXAMINATIONS COUNCIL
CARIBBEAN SECONDARY EDUCATION CERTIFICATE®
EXAMINATION
PHYSICS

Paper 01 – General Proficiency*1 hour 15 minutes***17 JANUARY 2013 (p.m.)****READ THE FOLLOWING INSTRUCTIONS CAREFULLY.**

1. This test consists of 60 items. You will have 1 hour and 15 minutes to answer them.
2. In addition to this test booklet, you should have an answer sheet.
3. Each item in this test has four suggested answers lettered (A), (B), (C), (D). Read each item you are about to answer and decide which choice is best.
4. On your answer sheet, find the number which corresponds to your item and shade the space having the same letter as the answer you have chosen. Look at the sample item below.

Sample Item

The SI unit of length is the

- (A) newton
(B) metre
(C) kilogram
(D) second

Sample Answer

(A) ☐ (B) ☒ (C) ☐ (D) ☐

5. The best answer to this item is “metre” so answer space (B) has been shaded.

If you want to change your answer, erase it completely before you fill in your new choice.

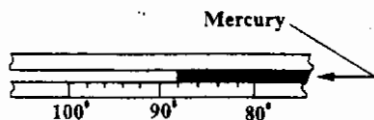
6. When you are told to begin, turn the page and work as quickly and as carefully as you can. If you cannot answer an item, go on to the next one. You may return to this item later. Your score will be the total number of correct answers.
7. You may do any rough work in this booklet.
8. Figures are not necessarily drawn to scale.
9. You may use a silent, non-programmable calculator to answer items.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

1. A physical quantity which is NOT one of the fundamental quantities is

(A) mass
(B) time
(C) current
(D) density

Item 5 refers to the following diagram which shows the section of a thermometer.



2. 0.0000357 N expressed in standard form is

(A) 3.57×10^{-5} N
(B) 3.75×10^4 N
(C) 3.75×10^{-4} N
(D) 3.75×10^5 N

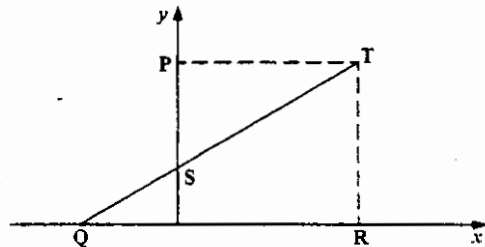
5. What is the reading shown?

(A) 88°
(B) 89°
(C) 91°
(D) 92°

3. The period of a simple pendulum is MOST affected by the

(A) length of the string
(B) amplitude of the oscillation
(C) mass of the bob
(D) force used to start it

Item 6 refers to the graph below which shows a relationship represented by the straight line passing through QT.



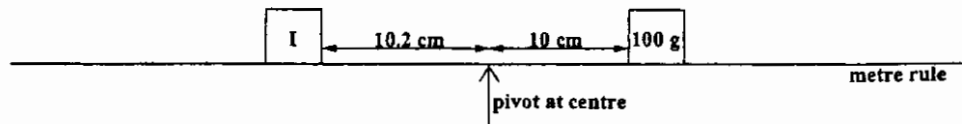
4. Which of the following is suitable for measuring the diameter of a human hair?

(A) Tape measure
(B) Vernier caliper
(C) Micrometer screw gauge
(D) Metre rule

6. For this relationship, when $x = 0$, the value of y is

(A) Q
(B) S
(C) P
(D) R

Item 7 refers to the diagram below.



7. The diagram above represents a 100 g mass which can be balanced by placing a mass at I. If the mass at I is to be used to balance the 100 g, it should be

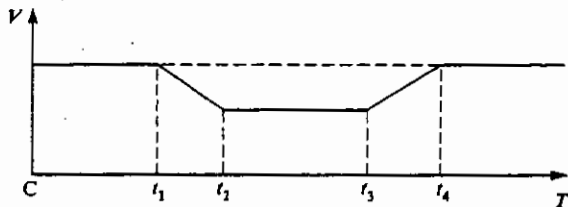
(A) less than 100 g
(B) 100 g
(C) a little greater than 100 g
(D) about 200 g

8. It is NOT true that the moment of a force is
- (A) measured in units called newton metres
- (B) the amount of force needed to keep a body turning
- (C) the turning effect that a force has when it acts on a body
- (D) the product of the force and the perpendicular distance of its line of action from a point
9. In an experiment to locate the centre of mass of a sheet of cardboard, a plumb line (string with a small mass on the end) is used. The plumb line is used to
- (A) check that the apparatus is vertical
- (B) check that the cardboard is swinging freely
- (C) show which points are vertically below the pivot
- (D) measure the width of the cardboard

10. Which of the following is a vector physical quantity?

- (A) Speed
- (B) Energy
- (C) Mass
- (D) Displacement

Item 11 refers to the following diagram which shows a velocity/time graph for a moving object.



11. Which of the following statements about the object is/are true?

- I. It returns to its starting point.
- II. It has zero acceleration between times t_2 and t_3 .
- III. Its velocity at t_4 is the same as its initial velocity.

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

12. Two smooth spheres, A and B, collide head on. Which of the following statements is/are true?

- I. The momentum of A is the same after collision as it was before.
- II. The momentum of B is the same after collision as it was before.
- III. The total momentum of A and B is the same after collision as it was before.

- (A) I only
- (B) III only
- (C) II and III only
- (D) I, II and III

13. Which of the following is NOT a vector quantity?

- (A) Displacement
- (B) Density
- (C) Acceleration
- (D) Momentum

14. Which of the following are renewable sources of energy?

- I. Solar energy
- II. Wind energy
- III. Geothermal energy

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

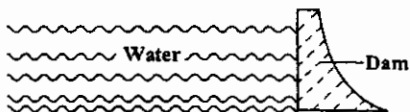
15. An ice cube sinks in liquid A but floats in liquid B. Which of the following statements is true?

- (A) The upthrust is less in A than in B.
- (B) The upthrust is greater in A than in B.
- (C) The weight of the ice is less in A than in B.
- (D) The weight of the ice is greater in A than in B.

16. A piece of string is tied onto a small stone and the stone is then suspended, totally immersed, in water. The tension in the string will be

- (A) zero
- (B) equal to the weight of the stone
- (C) less than the weight of the stone
- (D) more than the weight of the stone

Item 17 refers to the following diagram of a dam.



17. The diagram above shows a dam. The pressure on the dam at the bottom of the reservoir depends on the

- (A) depth of the water
- (B) volume of water held by the dam
- (C) mass of water held back by the dam
- (D) length of the reservoir

18. Which of the following tools is/are designed to take advantage of a large moment provided by a relatively small force?

- I. Claw hammer
- II. Crowbar
- III. Pair of tweezers
- IV. Pair of wire cutters

- (A) III only
- (B) I and IV only
- (C) I, II and III only
- (D) I, II and IV only

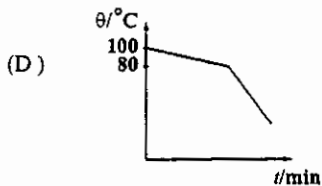
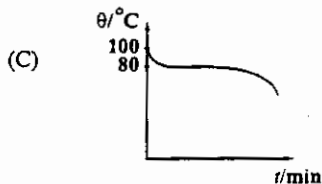
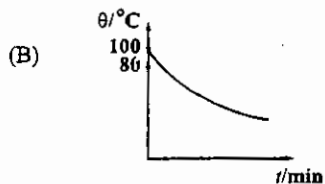
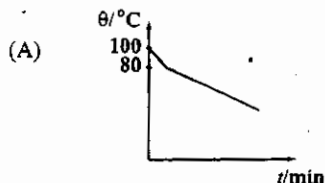
19. The temperature on a warm day in the Caribbean is about

(A) 32 K
(B) 240 K
(C) 273 K
(D) 305 K

20. Which of the following is MOST likely to account for the fact that pot handles are usually made of wood or plastic?

(A) Conduction
(B) Convection
(C) Radiation
(D) Evaporation

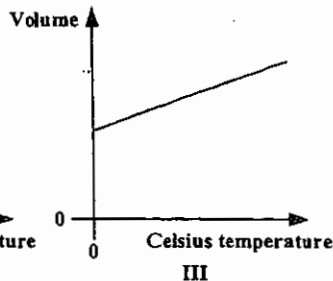
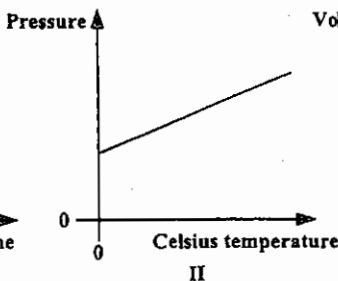
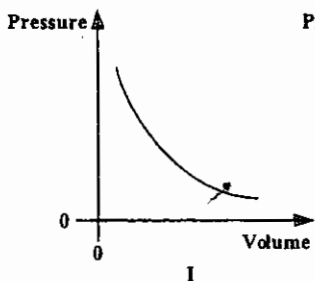
21. Some molten naphthalene at 100 °C is allowed to cool down to room temperature. If naphthalene has a melting point of 80 °C, which of the following graphs BEST represents the cooling curve?



22. Which of the following does NOT describe a process which is an example of evaporation?

(A) A slice of bread left in the open air becomes dry.
(B) The cooling effect of sweating in animals.
(C) The rapid disappearance of ether if exposed to the air.
(D) A loaded copper wire put around a block of ice gradually cuts through the ice.

Item 23 refers to the following diagrams which show how the pressure, the volume and the temperature of a fixed mass of dry gas, are inter-related.



23. Which of the graphs can be used to establish the kelvin temperature scale?

- (A) II only
- (B) I and II only
- (C) II and III only
- (D) I, II and III

24. The specific latent heat of vapourization of water is the energy required to change 1 kg of water at

- (A) 0 °C to steam at 100 °C
- (B) 0 °C to ice at 0 °C
- (C) 99.9 °C to steam at 100.1 °C
- (D) 100 °C to steam at 100 °C

25. An electric kettle full of water is plugged into the mains. The process by which heat travels through the water is

- (A) electrification
- (B) convection
- (C) evaporation
- (D) radiation

26. A detector of thermal energy is placed an equal distance in turn from each of four faces of a hollow metal cube full of water. The reading on the detector is GREATEST when the detector is turned towards the face which is

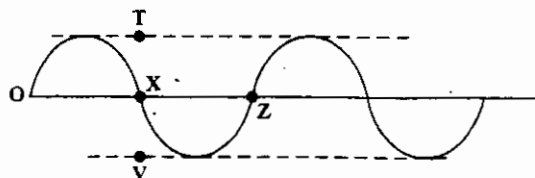
- (A) painted silver
- (B) painted dull black
- (C) painted shiny white
- (D) highly polished

27. Which of the following waves travel only longitudinally?

- I. Sound waves
- II. Radio waves
- III. Water waves

- (A) I only
- (B) II only
- (C) II and III only
- (D) I, II and III

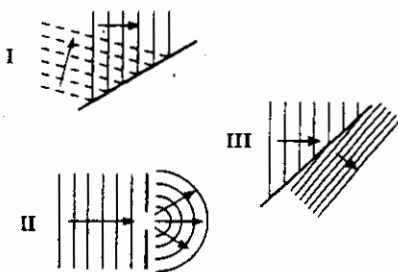
Item 28 refers to the following diagram which represents a wave travelling to the right.



28. The amplitude of the wave is the distance

- (A) OX
- (B) OZ
- (C) TV
- (D) TX

29. Which of the following diagrams could represent diffraction of water waves in a ripple tank?



- (A) I only
- (B) II only
- (C) I and II only
- (D) II and III only

Item 30 refers to the table below which lists the refractive indices for light of four different materials.

Material	Refractive Index
Air	1.0
Ice	1.3
Perspex	1.5
Diamond	2.4

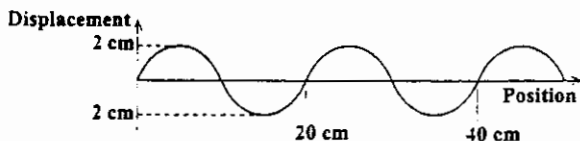
30. In which medium would the light waves have the SLOWEST speed?

- (A) Air
- (B) Ice
- (C) Perspex
- (D) Diamond

31. Young's double slit experiment demonstrates that light

- (A) has a particle nature
- (B) is a wave motion
- (C) travels in a straight line
- (D) produces sharp images

Item 32 refers to the diagram below which shows an instantaneous profile of a wave travelling across a water surface.



32. From the information given, the frequency is

- (A) $\frac{1}{20}$ Hz
- (B) 10 Hz
- (C) 20 Hz
- (D) unknown

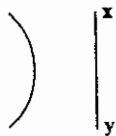
33. If a converging lens is used as a magnifying glass, the image formed is

(A) real, inverted, diminished
(B) real, erect, diminished
(C) virtual, inverted, magnified
(D) virtual, erect, magnified

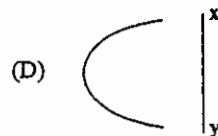
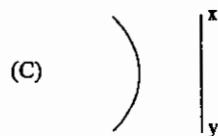
34. Which row in the table below correctly compares X rays and visible light?

	X- rays	Visible Light
(A)	Particles	Waves
(B)	High frequency	Low frequency
(C)	Long wavelength	Short wavelength
(D)	Carry less energy	Carry more energy

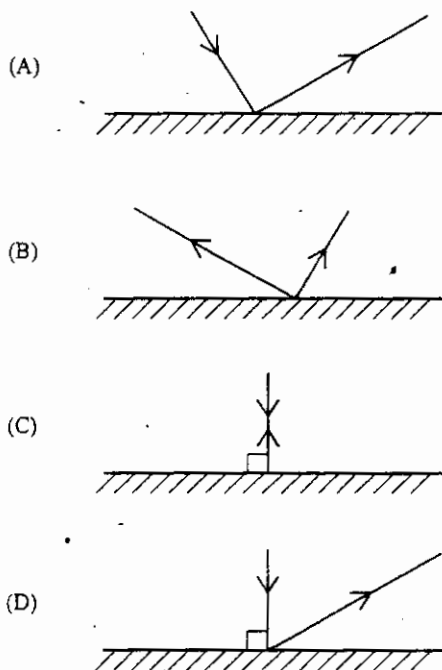
- Item 35 refers to the following diagram.



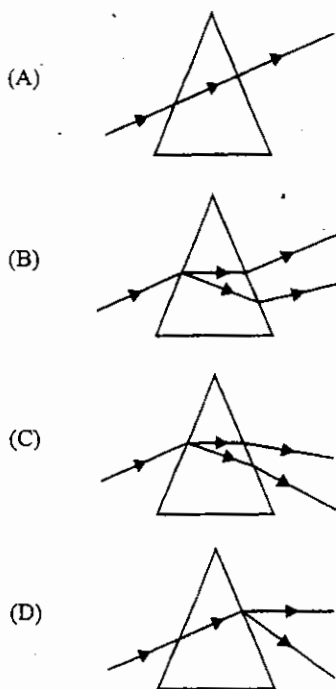
35. The circular wavefront shown above is progressing towards the reflecting surface xy. Which of the diagrams below correctly shows the shape of the wavefront after reflection?



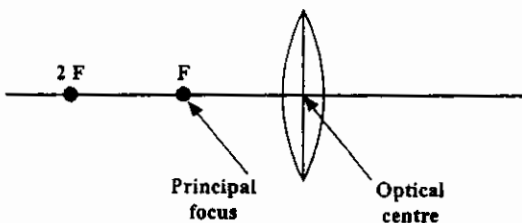
36. Which of the following diagrams MOST clearly shows the path of a ray of light when it strikes a plane mirror?



37. Which of the following diagrams BEST represents the passage of a beam of white light through a triangular glass prism?

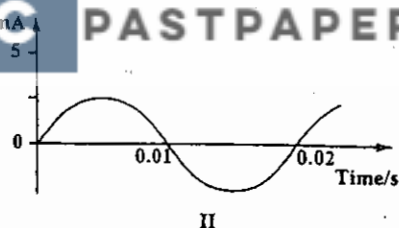
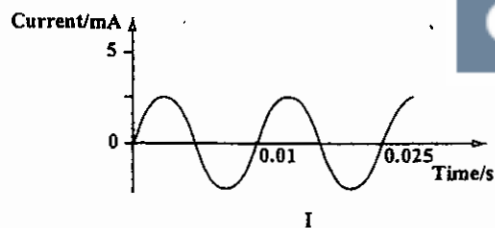


Item 38 refers to the diagram below.



38. With reference to the diagram above, a real image is produced with a converging lens when the object is located
- (A) between the optical centre and F
 - (B) between F and infinity
 - (C) at F only
 - (D) at 2F only

Item 39 refers to the graphs below which represent two alternating currents.



39. Which of the following combinations correctly identifies these currents?

	Current with greater frequency	Current with greater peak value
(A)	I	II
(B)	II	Neither
(C)	Neither	I
(D)	I	Neither

40. Which of the following statements about insulators is/are true?

- I. In an insulator all electrons are bound firmly to their atoms.
- II. In an insulator many electrons can move freely from atom to atom.
- III. An insulator cannot be charged by rubbing.
- IV. A good insulator retains the charge better than a conductor.

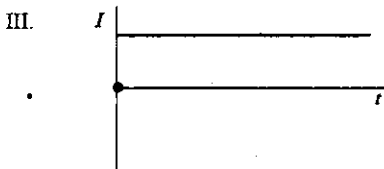
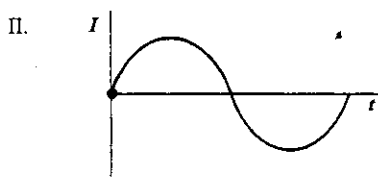
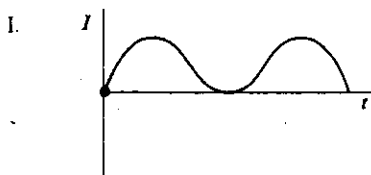
- (A) I only
- (B) I and II only
- (C) II and III only
- (D) II and IV

41. The current in a wire is one ampere if a charge of

- (A) 10 coulombs flows through it in 10 seconds
- (B) 1 coulomb flows through it in 1 second
- (C) 10 coulombs flow through it in 1 second
- (D) 100 coulombs flows through it in 10 seconds

Item 42 refers to the following current-time graphs.

Item 44 refers to the diagram below.



44. In the parallel circuit shown above, ALL of the current, I , supplied by the battery

- (A) goes through each of the resistors
- (B) goes through the largest resistor only
- (C) goes through the smallest resistor only
- (D) equals the sum of the currents in the two resistors

45. Which of the following relationships gives a correct value for the combined resistance R_T of resistors R_1 , R_2 , and R_3 connected in parallel?

- (A) $R_T = R_1 + R_2 + R_3$
- (B) $R_T = \frac{R_1 R_2 R_3}{R_1 + R_2 + R_3}$
- (C) $R_T = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$
- (D) $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$

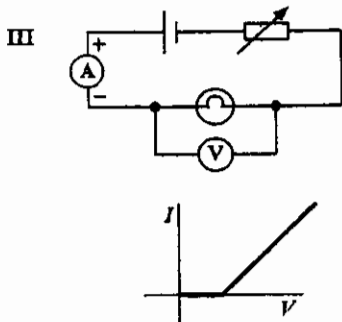
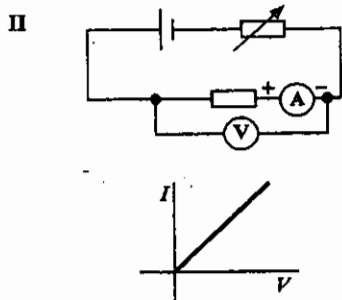
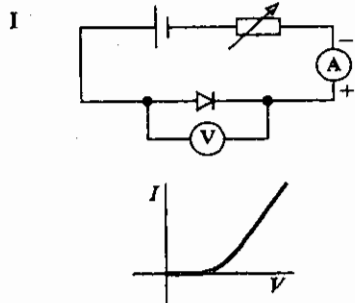
42. Which of the graphs above represent(s) direct current?

- (A) II only
- (B) III only
- (C) I and III only
- (D) I, II and III

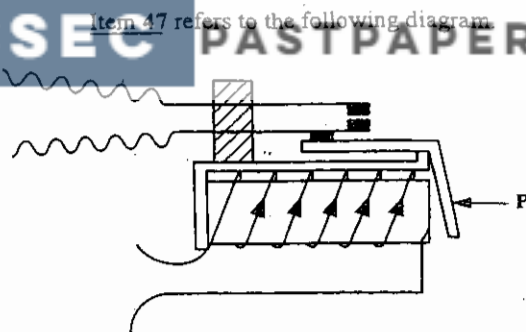
43. Which of the following statements about alternating current is true?

- (A) It can be changed into direct current by a transformer.
- (B) It can be rectified by using a semiconductor diode.
- (C) It can be used to recharge a battery.
- (D) It is used to transmit electrical energy because of its high frequency.

46. In I, II and III below, which graph could have been obtained from measurements made using the circuit above it?



- (A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III



47. The diagram above shows a typical relay. The part, labelled P, must be made of

- (A) brass
(B) iron
(C) plastic
(D) copper

48. Which of the following formulae (in which the symbols have their usual meaning) is a mathematical expression of Ohm's Law?

- (A) $V = IR$
(B) $P = VI$
(C) $Q = IT$
(D) $E = PT$

49. Rectification can BEST be done by using a

- (A) transformer
(B) capacitor
(C) transistor
(D) diode

50. Magnetic induction occurs when

- (A) a N pole attracts a S pole
(B) iron nails near a magnet become magnetized
(C) a magnet is suspended and points in the N S direction
(D) an electroscope is charged

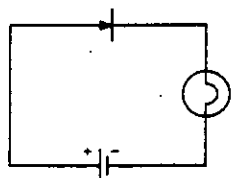
51. The direction of a magnetic field is the direction in which

- (A) the S pole of a magnet points when placed in it
- (B) the N pole of a magnet points when placed in it
- (C) a positive charge moves when placed in it
- (D) a negative charge moves when placed in it

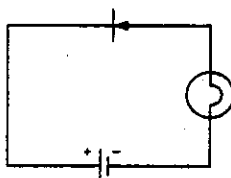
52. In which of the following devices is a commutator used?

- (A) Transformers
- (B) A C generators
- (C) D C motors
- (D) Moving coil microphones

Item 53 refers to the following circuit diagrams.



I



II

53. A simple experiment was conducted using the circuit diagrams shown above. The same components were used and the bulb was lit to normal brightness in each case.

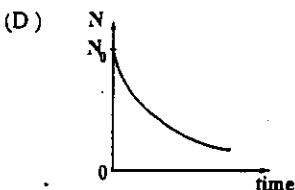
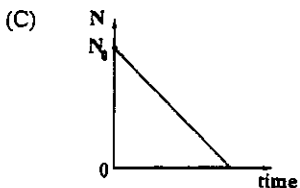
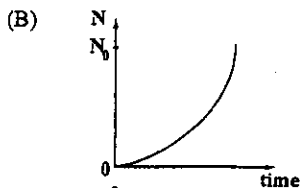
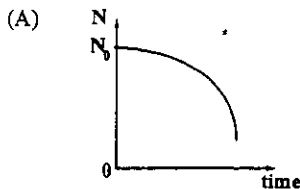
Which of the following components would be defective?

- I. The bulb
- II. The battery
- III. The diode

- (A) I only
- (B) III only
- (C) I and II only
- (D) II and III only

54. No radioactive nuclei are present in a sample at time $t = 0$.

Which of the following graphs BEST represents the variation with time of the number, N , of undecayed nuclei present?



55. Which of the following actions is NOT one of the ways in which the strength of the magnetic field near a solenoid (long coil) carrying a current can be increased?
- (A) Increasing the resistance of the coil
(B) Increasing the current in the coil
(C) Increasing the number of turns per unit length of the coil
(D) Placing a soft iron core inside the coil
56. Isotopes of an element differ from each other in that they have a different
- (A) number of electrons
(B) number of protons
(C) number of neutrons
(D) atomic number
57. Gamma rays may be stopped by
- (A) a sheet of paper
(B) a few millimetres of aluminium
(C) a few centimetres of aluminium
(D) several centimetres of lead
58. Which of the following are definitions of the term 'half-life' of radioactive nuclide?
- I. The time taken for the activity of any given sample to fall to half its original value.
II. The time taken for half the nuclei present in any given sample to decay.
III. Half the average number of disintegrations per second.
- (A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III
59. Which of the following pairs of terms describes two properties of an α -particle?
- (A) No charge, very penetrating
(B) Positive charge, very penetrating
(C) Positive charge, not very penetrating
(D) Negative charge, not very penetrating
60. In the equation $\Delta E = \Delta mc^2$
- (A) c = speed of light, Δm = mass of atom
(B) c = speed of light, Δm = change in mass
(C) c = specific heat capacity of substance, Δm = change in mass
(D) c = specific heat capacity of substance, Δm = mass of substance

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.